

Pritam

Portfolio: prtmxio

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Education

Deenbandhu Chhotu Ram University of Science and Technology

2023 - Present

Bachelor of Technology in Electronics & Communication Engineering; CGPA: 8.43/10

Sonapat, Haryana

CBSE - 12

2022

Mathematics - Grade: 96.2%

New Delhi

Experience

Embedded Application Developer (Intern)

Jul 2025 – Dec 2025

PCB CUPID, Bangalore

Remote

- Helped build and launch **Circuit Spell** (spell.pcbcupid.com) from ideation – an AI agent that converts natural language prompts into full PCB designs using 50+ proprietary Glyph modules, generating schematics, routed layouts, and 3D views exportable in Gerber/KiCad/PDF.
- Prototyped core EDA subsystems – constraint-based component placement and auto-routing – de-risking the architecture before the production build.
- Engineered multi-sensor IoT firmware in C/C++ on Glyph boards (C2, C6, H2) with interrupt-driven scheduling for high-precision industrial control tasks.

Robotics Engineering Team Member and Mentor

Oct 2023 – Present

THINKBOTS Robotics Club, DCRUST

Murthal, Haryana

- Architected and built autonomous line-following robots using Arduino, achieving 95% path accuracy on complex competition tracks.
- Led and mentored a team of 15+ junior developers, teaching best practices in embedded systems, PID control, and algorithm design.

Projects

XIO-SAT – Fault-Tolerant CubeSat Guardian Node | ESP-IDF, ForgeFPGA, ROS 2, MicroPython

Feb 2025

- Engineered an SLG47910 FPGA watchdog over a 4-node stack (FPGA + RP2040 + ESP32-C6 + MPU6050) monitoring a 5Hz heartbeat, guaranteeing hard reboot within **1.5s** of fault with reset cause persisted to NVS flash across power cycles.
- Built an ADCS telemetry pipeline: MPU6050 → RP2040 complementary filter (20Hz, 200-sample calibration) → UART → ESP32-C6 → MQTT → ROS 2; ground-triggered fault injection validates FPGA watchdog recovery end-to-end in under **2s**.
- Architected a ROS 2 ↔ MQTT bridge multiplexing **3 live telemetry streams at 20Hz** with bidirectional command relay for kill-injection, boot forensics, and anomaly reporting on a 1U form factor.

Synapse - Local Multimodal Perceive-Retrieve-Generate Pipeline | Transformers, Diffusers, HNSW | 🐙 Mar 2026

- Engineered a 3-stage offline pipeline chaining **SmolVLM-500M** (VLM captioning), **CLIP ViT-B/32** (contrastive retrieval), and **SD-Turbo** (single-step diffusion) within a hard **4 GB VRAM** budget via staged fp16 model loading and explicit GPU memory handoff.
- Built CLIP-based semantic image retrieval using **HNSW approximate nearest-neighbour indexing** (cosine space, 512-dim), achieving **~1ms query time** over 5,000+ indexed images vs. O(N) brute-force re-encoding.
- Reduced end-to-end generation latency to **~6s** (prompt mode) and **~12s** (image mode) on RTX 2050 by co-loading compatible models and skipping the VLM stage when not required.

Technical Skills

Languages: C/C++, Python, Java, Verilog

Embedded & Robotics: ESP-IDF, ROS 2, Arduino, ForgeFPGA, MQTT, PID Control, Multithreading, Microcontroller Firmware

AI / ML: PyTorch, HuggingFace, Transformers, CNN, Diffusers, OpenCV, Scikit-learn

Tools: Git, Linux, KiCad, VS Code, Vim

Achievements & Recognition

Smart India Hackathon - National Level Competition — *Team Leader & Software Developer*

2024

- Built a real-time AI face recognition system for missing persons identification, deployed at Simhastha Ujjain Mela in collaboration with the Police Department.